



Qualification Pack



Mushroom Cultivator

QP Code: WBSC/AGR/Q3101

Version: 1.0

NSQF Level: 3

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WBSC/AGR/Q3101: Mushroom Cultivator

Brief Job Description

A mushroom cultivator primarily carries out ex-situ cultivation of mushrooms. Perform routine care including planting, irrigating and harvesting of mushrooms. Examine quality of mushroom produce and prepare items for market/sale purposes. Monitor mushroom growing conditions including set water lines, air flow and barn temperature. Ensure the cleanness of the farm working.

Personal Attributes

A mushroom cultivator is hard working, focused and safety oriented. They observe the crop closely to notice growth changes, solve problems early and keep the work area clean and hygienic. They have the physical strength to carry out tasks like planting, watering and harvesting. They are responsible and disciplined and follow proper cultivation methods and can adjust to changes in the environment.

Applicable National Occupational Standards (NOS)

Compulsory NOS:

1. [WBSC/AGR/N3101: Identify the Scope & importance of Mushroom cultivation in India.](#)
2. [WBSC/AGR/N3102: Prepare & pasteurize the compost necessary to cultivate mushrooms.](#)
3. [WBSC/AGR/N3103: Select commercially important species of mushroom and design appropriate site to cultivate mushrooms.](#)
4. [WBSC/AGR/N3104: Undertake disease control and pest management activities, casing and pinning for mushroom cultivation.](#)
5. [WBSC/AGR/N3105: Demonstrate harvest & post-harvest procedures of mushrooms.](#)
6. [WBSC/AGR/N3106: Work in real job situation with special emphasis on basic safety and hazards in this domain \(Mushroom Cultivator\).](#)
7. [DGT/VSQ/N0102: Employability Skills \(60 Hours\)](#)

Qualification Pack (QP) Parameters

Sector	Agriculture
Sub-Sector	Other Allied
Occupation	Mushroom Cultivator



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Country	India
NSQF Level	3
Credits	12
Aligned to NCO/ISCO/ISIC Code	6194.9900
Minimum Educational Qualification & Experience	10th Class OR 8th grade pass and pursuing continuous schooling in regular school with vocational subject OR 8th Class pass with 2 Years of experience OR 5th grade pass with 5 Years of experience
Minimum Level of Education for Training in School	
Pre-Requisite License or Training	NA
Minimum Job Entry Age	18 Years
Last Reviewed On	NA
Next Review Date	03/05/2026
NSQC Approval Date	03/05/2023
Version	1.0
Reference code on NQR	QG-03-AG-00365-2023-V2-WBS
NQR Version	2



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WBSC/AGR/N3101: Identify the Scope & importance of Mushroom cultivation in India.

Description

Identify the Scope & importance of Mushroom cultivation in India.

Scope

The scope covers the following :

- The scope and importance of cultivation of mushroom in India scenario.
- The market demand of mushroom.
- The role of mushroom cultivator.

Elements and Performance Criteria

To be competent, the user/individual on the job must be able to:

PC1. Identify the scope and importance of cultivation of mushroom in India scenario.

PC2. Identify the market demand of mushroom.

PC3. Identify the role of mushroom cultivator.

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

KU1. The Scope & importance of Mushroom cultivation in India.

KU2. The usage & market demand for mushroom.

KU3. The Role of a Mushroom Grower.

Generic Skills (GS)

User/individual on the job needs to know how to:

GS1. Identify the scope and importance of cultivation of mushroom in India scenario.

GS2. Identify the market demand of mushroom.

GS3. Identify the role of mushroom cultivator.



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Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
	50	40	-	-
PC1. Identify the scope and importance of cultivation of mushroom in India scenario.	-	-	-	-
PC2. Identify the market demand of mushroom.	-	-	-	-
PC3. Identify the role of mushroom cultivator.	-	-	-	-
NOS Total	50	40	-	-



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National Occupational Standards (NOS) Parameters

NOS Code	WBSC/AGR/N3101
NOS Name	Identify the Scope & importance of Mushroom cultivation in India.
Sector	Agriculture
Sub-Sector	
Occupation	Mushroom Cultivator
NSQF Level	3
Credits	1
Version	1.0
Last Reviewed Date	03/05/2023
Next Review Date	03/05/2026
NSQC Clearance Date	03/05/2023



Qualification Pack

WBSC/AGR/N3102: Prepare & pasteurize the compost necessary to cultivate mushrooms.

Description

Prepare & pasteurize the compost necessary to cultivate mushrooms.

Scope

The scope covers the following :

- Appropriate materials to prepare the compost.
- Various agricultural by-products, materials rich in cellulose.
- Selection & application of chemicals for mineral deficiency rectification and stabilization.
- Different types and formulation of compost.
- Selection of composting methods.
- Compost rotation.
- Pasteurization of the compost.
- Good compost attributes and role of composting in Mushroom cultivation.
- Different types of compost.
- Determination of quality of compost.
- Hazards & risks associated with composting and how to control injury to self.

Elements and Performance Criteria

To be competent, the user/individual on the job must be able to:

- PC1.** Identify and Select appropriate materials to prepare the compost- base materials.
- PC2.** Describe various agricultural by-products, materials rich in cellulose.
- PC3.** Select & apply chemicals for mineral deficiency rectification and stabilization.
- PC4.** Identify different types compost- natural & synthetic, formulation of different compost.
- PC5.** Select composting methods- short, long; indoor, outdoor.
- PC6.** Undertake compost rotation and ensure adequate moisture, carbohydrate, gas exchange etc.
- PC7.** Pasteurize the compost to kill insects, nematodes, pest fungi or other pests.
- PC8.** Explore good compost attributes role of composting in Mushroom cultivation.
- PC9.** Prepare different types of compost by selecting appropriate materials.
- PC10.** Determine of quality of compost.
- PC11.** Identify hazards & risks associated with composting and how to control injury to self.

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** The appropriate materials to prepare the compost- base materials from various agricultural by-products, materials rich in cellulose.



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- KU2.** Proper physical structure to the substrate, ensure adequate aeration during composting, and adding bulk to the compost.
- KU3.** Different types compost- natural & synthetic, formulation of different compost.
- KU4.** Composting methods- short, long, indoor, outdoor.
- KU5.** Compost rotation and ensuring adequate moisture, carbohydrate, gas exchange etc.
- KU6.** Good compost attributes.

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** Demonstrate the role of composting in Mushroom cultivation.
- GS2.** Identify appropriate materials to prepare different types of compost.
- GS3.** Select & apply chemicals for mineral deficiency rectification and stabilization.
- GS4.** Demonstrate the methods of composting - preparation and pasteurization.
- GS5.** Pasteurize the compost to kill insects, nematodes, pest fungi, or other pests.
- GS6.** Determine the quality of compost.
- GS7.** Identify the hazards & risks associated with composting and how to control injury to self.



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Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
	24	60	-	-
PC1. Identify and Select appropriate materials to prepare the compost- base materials.	-	-	-	-
PC2. Describe various agricultural by-products, materials rich in cellulose.	-	-	-	-
PC3. Select & apply chemicals for mineral deficiency rectification and stabilization.	-	-	-	-
PC4. Identify different types compost- natural & synthetic, formulation of different compost.	-	-	-	-
PC5. Select composting methods- short, long; indoor, outdoor.	-	-	-	-
PC6. Undertake compost rotation and ensure adequate moisture, carbohydrate, gas exchange etc.	-	-	-	-
PC7. Pasteurize the compost to kill insects, nematodes, pest fungi or other pests.	-	-	-	-
PC8. Explore good compost attributes role of composting in Mushroom cultivation.	-	-	-	-
PC9. Prepare different types of compost by selecting appropriate materials.	-	-	-	-
PC10. Determine of quality of compost.	-	-	-	-
PC11. Identify hazards & risks associated with composting and how to control injury to self.	-	-	-	-
NOS Total	24	60	-	-



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National Occupational Standards (NOS) Parameters

NOS Code	WBSC/AGR/N3102
NOS Name	Prepare & pasteurize the compost necessary to cultivate mushrooms.
Sector	Agriculture
Sub-Sector	
Occupation	Mushroom Cultivator
NSQF Level	3
Credits	1
Version	1.0
Last Reviewed Date	03/05/2023
Next Review Date	03/05/2026
NSQC Clearance Date	03/05/2023



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WBSC/AGR/N3103: Select commercially important species of mushroom and design appropriate site to cultivate mushrooms.

Description

Select commercially important species of mushroom and design appropriate site to cultivate mushrooms.

Scope

The scope covers the following :

- Commercially important types of mushroom based on market demand.
- Procurement of mushroom spawns from authentic source.
- Selection of appropriate mushroom cultivation site with proper drainage & water supply facility.
- Designing and construction mushroom farm.
- Different types of mushroom growing facilities and fixtures.
- Types, components and their specifications of bulk chamber conducive for good quality mushroom growing.
- Package of practices of White button Mushroom and Oyster Mushroom.

Elements and Performance Criteria

To be competent, the user/individual on the job must be able to:

PC1. Select commercially important types of mushroom based on market's demand.

PC2. Select of important types of Mushroom.

PC3. Procure mushroom spawns from authentic source.

PC4. Select appropriate mushroom cultivation site with proper drainage & water supply facility.

PC5. Design and construct mushroom farm according to the growing conditions required for different kinds of mushrooms.

PC6. Understand different types of mushroom growing facilities and fixtures.

PC7. Understand types, components and their specifications of bulk chamber conducive for good quality mushroom growing.

PC8. Package of practices of White button Mushroom and Oyster Mushroom.

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

KU1. Selection of commercially important type of mushroom based on markets demand, climatic conditions of the farm, growing season, investments, etc.

KU2. Selection of appropriate mushroom cultivation site with proper drainage & water supply facility.

KU3. Designing and construction of mushroom farm according to the growing conditions required for different kinds of mushrooms.

KU4. Different types of mushroom growing facilities and fixtures.



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KU5. Types, components and their specifications of bulk chamber conducive for good quality mushroom growing.

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** Select important types of Mushroom.
- GS2.** Select appropriate Mushroom cultivation sites.
- GS3.** Procure mushroom spawns from authentic source.
- GS4.** Design and construct Mushroom farm.



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Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
	26	60	-	-
PC1. Select commercially important types of mushroom based on market's demand.	-	-	-	-
PC2. Select of important types of Mushroom.	-	-	-	-
PC3. Procure mushroom spawns from authentic source.	-	-	-	-
PC4. Select appropriate mushroom cultivation site with proper drainage & water supply facility.	-	-	-	-
PC5. Design and construct mushroom farm according to the growing conditions required for different kinds of mushrooms.	-	-	-	-
PC6. Understand different types of mushroom growing facilities and fixtures.	-	-	-	-
PC7. Understand types, components and their specifications of bulk chamber conducive for good quality mushroom growing.	-	-	-	-
PC8. Package of practices of White button Mushroom and Oyster Mushroom.	-	-	-	-
NOS Total	26	60	-	-



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National Occupational Standards (NOS) Parameters

NOS Code	WBSC/AGR/N3103
NOS Name	Select commercially important species of mushroom and design appropriate site to cultivate mushrooms.
Sector	Agriculture
Sub-Sector	
Occupation	Mushroom Cultivator
NSQF Level	3
Credits	1
Version	1.0
Last Reviewed Date	03/05/2023
Next Review Date	03/05/2026
NSQC Clearance Date	03/05/2023



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WBSC/AGR/N3104: Undertake disease control and pest management activities, casing and pinning for mushroom cultivation.

Description

Undertake disease control and pest management activities, casing and pinning for mushroom cultivation.

Scope

The scope covers the following :

- Inspection of mushroom bags or beds.
- Identification of the diseases.
- Controlling diseases and exercising preventive care.
- Pasteurization of the mushroom farm to remove nematode in mushroom cultivation.
- Preparation for casing soil to hold moisture.
- Promotion of the formation of primordia, or mushroom pins by supplying water to the mycelium.
- Detection of the earliest formation of recognizable mushrooms from mycelium.
- Usage of sterilized casing to control nematodes.
- Spraying fungicide after casing to check dry bubble.
- Spraying insecticide for control of mites.
- Application caustic chemicals top keep rodents away.

Elements and Performance Criteria

To be competent, the user/individual on the job must be able to:

- PC1.** Inspect mushroom bags or beds carefully for early detection of pests and diseases.
- PC2.** Identify the diseases.
- PC3.** Control diseases and exercise preventive care- spray pesticides/ fungicides etc.
- PC4.** Pasteurize the mushroom farm to remove nematode in mushroom cultivation.
- PC5.** Prepare casing soil to hold moisture.
- PC6.** Promote the formation of primordia, or mushroom pins by supplying water to the mycelium.
- PC7.** Detect the earliest formation of recognizable mushrooms from mycelium.
- PC8.** Use sterilized casing to control nematodes.
- PC9.** Spray fungicide after casing to check dry bubble.
- PC10.** Spray insecticide for control of mites.
- PC11.** Apply caustic chemicals top keep rodents away.

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** Inspection of mushroom bags or beds carefully for early detection of pests and diseases.
- KU2.** Controlling of diseases and exercising preventive care- spraying pesticides/ fungicides etc.



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- KU3.** Pasteurization of the mushroom farm to remove nematode in mushroom cultivation.
- KU4.** Preparation of casing soil to hold moisture.
- KU5.** The formation of primordia, or mushroom pins by supplying water to the mycelium.
- KU6.** Detection of the earliest formation of recognizable mushrooms from mycelium case at a regular interval after harvesting or covering the holes after mushroom picking.

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** Inspect Mushroom bags or beds for early detection of pests and diseases.
- GS2.** Use sterilized casing to control nematodes.
- GS3.** Spray fungicide after casing to check dry bubble.
- GS4.** Spray insecticide for control of mites.
- GS5.** Use caustic chemicals to keep rodents away.



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Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
	24	60	-	-
PC1. Inspect mushroom bags or beds carefully for early detection of pests and diseases.	-	-	-	-
PC2. Identify the diseases.	-	-	-	-
PC3. Control diseases and exercise preventive care- spray pesticides/ fungicides etc.	-	-	-	-
PC4. Pasteurize the mushroom farm to remove nematode in mushroom cultivation.	-	-	-	-
PC5. Prepare casing soil to hold moisture.	-	-	-	-
PC6. Promote the formation of primordia, or mushroom pins by supplying water to the mycelium.	-	-	-	-
PC7. Detect the earliest formation of recognizable mushrooms from mycelium.	-	-	-	-
PC8. Use sterilized casing to control nematodes.	-	-	-	-
PC9. Spray fungicide after casing to check dry bubble.	-	-	-	-
PC10. Spray insecticide for control of mites.	-	-	-	-
PC11. Apply caustic chemicals top keep rodents away.	-	-	-	-
NOS Total	24	60	-	-



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National Occupational Standards (NOS) Parameters

NOS Code	WBSC/AGR/N3104
NOS Name	Undertake disease control and pest management activities, casing and pinning for mushroom cultivation.
Sector	Agriculture
Sub-Sector	
Occupation	Mushroom Cultivator
NSQF Level	3
Credits	1
Version	1.0
Last Reviewed Date	03/05/2023
Next Review Date	03/05/2026
NSQC Clearance Date	03/05/2023



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WBSC/AGR/N3105: Demonstrate harvest & post-harvest procedures of mushrooms.

Description

Demonstrate harvest & post-harvest procedures of mushrooms.

Scope

The scope covers the following :

- Assessment of the maturity of mushroom and harvest periods.
- Good harvesting practices.
- Cutting, cleaning and drying harvested mushroom using approved procedures.
- Sorting and grading the harvests as per required quality specifications.
- Storing, packaging, labelling and transporting produce.
- Recording the information in appropriate registers.
- Recording book and logs.
- Utilization of spent mushroom substrate in organic farming, vermi composting, bioremediation of contaminated soil etc.
- Sorting of the Mushrooms on the size and quality.
- Packaging Mushrooms with labels.
- Usage of spent Mushroom in vermi-composting and in organic farming.

Elements and Performance Criteria

To be competent, the user/individual on the job must be able to:

- PC1.** Assess the maturity of mushroom and harvest periods.
- PC2.** Apply good harvesting practices.
- PC3.** Cut, clean and dry harvested mushroom using approved procedures.
- PC4.** Sort and grade the harvests as per required quality specifications.
- PC5.** Store, pack, label and transport produce.
- PC6.** Record the information, e.g. quality, quantity, type, expenditure incurred in operation, etc. in appropriate registers, record book and logs.
- PC7.** Utilize spent mushroom substrate in organic farming, vermi composting, bioremediation of contaminated soil etc.
- PC8.** Sorting the Mushrooms on the size and quality.
- PC9.** Packaging Mushrooms with labels containing month and year of harvesting, quantity and type of Mushroom etc.
- PC10.** Use of spent Mushroom in vermi-composting and in organic farming.

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:



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- KU1.** Assessment of the maturity of mushroom and harvest periods.
- KU2.** Application of good harvesting practices.
- KU3.** Cutting, cleaning and dry harvesting mushroom using approved procedures.
- KU4.** Sorting and grading of the harvests as per required quality specifications.
- KU5.** Storing, packing, labelling and transporting of produce.
- KU6.** Recording of information, e.g. quality, quantity, type, expenditure incurred in operation, etc. in appropriate registers, recording book and logs.
- KU7.** Utilization of spent mushroom substrate in organic farming, vermicomposting, bioremediation of contaminated soil etc.

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** Identify right stage of Mushroom.
- GS2.** Identify methods of harvesting.
- GS3.** Use approved cutting techniques for harvesting.
- GS4.** Sort the mushrooms on the size and quality.
- GS5.** Pack Mushrooms with labels containing month and year of harvesting, quantity and type of Mushroom etc.
- GS6.** Use spent Mushroom in vermi-composting and in organic farming.
- GS7.** Prepare value added products of Mushroom.



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Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
	26	140	-	-
PC1. Assess the maturity of mushroom and harvest periods.	-	-	-	-
PC2. Apply good harvesting practices.	-	-	-	-
PC3. Cut, clean and dry harvested mushroom using approved procedures.	-	-	-	-
PC4. Sort and grade the harvests as per required quality specifications.	-	-	-	-
PC5. Store, pack, label and transport produce.	-	-	-	-
PC6. Record the information, e.g. quality, quantity, type, expenditure incurred in operation, etc. in appropriate registers, record book and logs.	-	-	-	-
PC7. Utilize spent mushroom substrate in organic farming, vermi composting, bioremediation of contaminated soil etc.	-	-	-	-
PC8. Sorting the Mushrooms on the size and quality.	-	-	-	-
PC9. Packaging Mushrooms with labels containing month and year of harvesting, quantity and type of Mushroom etc.	-	-	-	-
PC10. Use of spent Mushroom in vermi-composting and in organic farming.	-	-	-	-
NOS Total	26	140	-	-



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National Occupational Standards (NOS) Parameters

NOS Code	WBSC/AGR/N3105
NOS Name	Demonstrate harvest & post-harvest procedures of mushrooms.
Sector	Agriculture
Sub-Sector	
Occupation	Mushroom Cultivator
NSQF Level	3
Credits	2
Version	1.0
Last Reviewed Date	03/05/2023
Next Review Date	03/05/2026
NSQC Clearance Date	03/05/2023



Qualification Pack

WBSC/AGR/N3106: Work in real job situation with special emphasis on basic safety and hazards in this domain (Mushroom Cultivator).

Description

Work in real job situation with special emphasis on basic safety and hazards in this domain.

Scope

The scope covers the following :

- Work in the real job situation.
- Report preparation on the competency developed.

Elements and Performance Criteria

To be competent, the user/individual on the job must be able to:

- PC1.** Work in the real job situation with special emphasis on basic safety and hazards in this domain.
- PC2.** Prepare report for this component of training of the course to ensure whether competency has been developed.

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** Work in the real job situation with special emphasis on basic safety and hazards in this domain.
- GS2.** Prepare report for this component of training of the course to ensure whether competency has been developed.



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Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
	-	440	-	-
PC1. Work in the real job situation with special emphasis on basic safety and hazards in this domain.	-	-	-	-
PC2. Prepare report for this component of training of the course to ensure whether competency has been developed.	-	-	-	-
NOS Total	-	440	-	-



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National Occupational Standards (NOS) Parameters

NOS Code	WBSC/AGR/N3106
NOS Name	Work in real job situation with special emphasis on basic safety and hazards in this domain (Mushroom Cultivator).
Sector	Agriculture
Sub-Sector	
Occupation	Mushroom Cultivator
NSQF Level	3
Credits	4
Version	1.0
Last Reviewed Date	03/05/2023
Next Review Date	03/05/2026
NSQC Clearance Date	03/05/2023



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DGT/VSQ/N0102: Employability Skills (60 Hours)

Description

This unit is about employability skills, Constitutional values, becoming a professional in the 21st Century, digital, financial, and legal literacy, diversity and Inclusion, English and communication skills, customer service, entrepreneurship, and apprenticeship, getting ready for jobs and career development.

Scope

The scope covers the following :

- Introduction to Employability Skills
- Constitutional values - Citizenship
- Becoming a Professional in the 21st Century
- Basic English Skills
- Career Development & Goal Setting
- Communication Skills
- Diversity & Inclusion
- Financial and Legal Literacy
- Essential Digital Skills
- Entrepreneurship
- Customer Service
- Getting ready for Apprenticeship & Jobs

Elements and Performance Criteria

Introduction to Employability Skills

To be competent, the user/individual on the job must be able to:

- PC1.** identify employability skills required for jobs in various industries
- PC2.** identify and explore learning and employability portals

Constitutional values – Citizenship

To be competent, the user/individual on the job must be able to:

- PC3.** recognize the significance of constitutional values, including civic rights and duties, citizenship, responsibility towards society etc. and personal values and ethics such as honesty, integrity, caring and respecting others, etc.
- PC4.** follow environmentally sustainable practices

Becoming a Professional in the 21st Century

To be competent, the user/individual on the job must be able to:

- PC5.** recognize the significance of 21st Century Skills for employment
- PC6.** practice the 21st Century Skills such as Self-Awareness, Behaviour Skills, time management, critical and adaptive thinking, problem-solving, creative thinking, social and cultural awareness, emotional awareness, learning to learn for continuous learning etc. in personal and professional life

Basic English Skills

To be competent, the user/individual on the job must be able to:



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- PC7.** use basic English for everyday conversation in different contexts, in person and over the telephone
- PC8.** read and understand routine information, notes, instructions, mails, letters etc. written in English
- PC9.** write short messages, notes, letters, e-mails etc. in English

Career Development & Goal Setting

To be competent, the user/individual on the job must be able to:

- PC10.** understand the difference between job and career
- PC11.** prepare a career development plan with short- and long-term goals, based on aptitude

Communication Skills

To be competent, the user/individual on the job must be able to:

- PC12.** follow verbal and non-verbal communication etiquette and active listening techniques in various settings
- PC13.** work collaboratively with others in a team

Diversity & Inclusion

To be competent, the user/individual on the job must be able to:

- PC14.** communicate and behave appropriately with all genders and PwD
- PC15.** escalate any issues related to sexual harassment at workplace according to POSH Act

Financial and Legal Literacy

To be competent, the user/individual on the job must be able to:

- PC16.** select financial institutions, products and services as per requirement
- PC17.** carry out offline and online financial transactions, safely and securely
- PC18.** identify common components of salary and compute income, expenses, taxes, investments etc
- PC19.** identify relevant rights and laws and use legal aids to fight against legal exploitation

Essential Digital Skills

To be competent, the user/individual on the job must be able to:

- PC20.** operate digital devices and carry out basic internet operations securely and safely
- PC21.** use e-mail and social media platforms and virtual collaboration tools to work effectively
- PC22.** use basic features of word processor, spreadsheets, and presentations

Entrepreneurship

To be competent, the user/individual on the job must be able to:

- PC23.** identify different types of Entrepreneurship and Enterprises and assess opportunities for potential business through research
- PC24.** develop a business plan and a work model, considering the 4Ps of Marketing Product, Price, Place and Promotion
- PC25.** identify sources of funding, anticipate, and mitigate any financial/ legal hurdles for the potential business opportunity

Customer Service

To be competent, the user/individual on the job must be able to:

- PC26.** identify different types of customers
- PC27.** identify and respond to customer requests and needs in a professional manner.



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PC28. follow appropriate hygiene and grooming standards

Getting ready for apprenticeship & Jobs

To be competent, the user/individual on the job must be able to:

PC29. create a professional Curriculum vitae (Résumé)

PC30. search for suitable jobs using reliable offline and online sources such as Employment exchange, recruitment agencies, newspapers etc. and job portals, respectively

PC31. apply to identified job openings using offline /online methods as per requirement

PC32. answer questions politely, with clarity and confidence, during recruitment and selection

PC33. identify apprenticeship opportunities and register for it as per guidelines and requirements

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

KU1. need for employability skills and different learning and employability related portals

KU2. various constitutional and personal values

KU3. different environmentally sustainable practices and their importance

KU4. Twenty first (21st) century skills and their importance

KU5. how to use English language for effective verbal (face to face and telephonic) and written communication in formal and informal set up

KU6. importance of career development and setting long- and short-term goals

KU7. about effective communication

KU8. POSH Act

KU9. Gender sensitivity and inclusivity

KU10. different types of financial institutes, products, and services

KU11. how to compute income and expenditure

KU12. importance of maintaining safety and security in offline and online financial transactions

KU13. different legal rights and laws

KU14. different types of digital devices and the procedure to operate them safely and securely

KU15. how to create and operate an e- mail account and use applications such as word processors, spreadsheets etc.

KU16. how to identify business opportunities

KU17. types and needs of customers

KU18. how to apply for a job and prepare for an interview

KU19. apprenticeship scheme and the process of registering on apprenticeship portal

Generic Skills (GS)

User/individual on the job needs to know how to:

GS1. read and write different types of documents/instructions/correspondence

GS2. communicate effectively using appropriate language in formal and informal settings



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- GS3.** behave politely and appropriately with all
- GS4.** how to work in a virtual mode
- GS5.** perform calculations efficiently
- GS6.** solve problems effectively
- GS7.** pay attention to details
- GS8.** manage time efficiently
- GS9.** maintain hygiene and sanitization to avoid infection



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Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Introduction to Employability Skills</i>	1	1	-	-
PC1. identify employability skills required for jobs in various industries	-	-	-	-
PC2. identify and explore learning and employability portals	-	-	-	-
<i>Constitutional values – Citizenship</i>	1	1	-	-
PC3. recognize the significance of constitutional values, including civic rights and duties, citizenship, responsibility towards society etc. and personal values and ethics such as honesty, integrity, caring and respecting others, etc.	-	-	-	-
PC4. follow environmentally sustainable practices	-	-	-	-
<i>Becoming a Professional in the 21st Century</i>	2	4	-	-
PC5. recognize the significance of 21st Century Skills for employment	-	-	-	-
PC6. practice the 21st Century Skills such as Self-Awareness, Behaviour Skills, time management, critical and adaptive thinking, problem-solving, creative thinking, social and cultural awareness, emotional awareness, learning to learn for continuous learning etc. in personal and professional life	-	-	-	-
<i>Basic English Skills</i>	2	3	-	-
PC7. use basic English for everyday conversation in different contexts, in person and over the telephone	-	-	-	-
PC8. read and understand routine information, notes, instructions, mails, letters etc. written in English	-	-	-	-
PC9. write short messages, notes, letters, e-mails etc. in English	-	-	-	-
<i>Career Development & Goal Setting</i>	1	2	-	-



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Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC10. understand the difference between job and career	-	-	-	-
PC11. prepare a career development plan with short- and long-term goals, based on aptitude	-	-	-	-
<i>Communication Skills</i>	2	2	-	-
PC12. follow verbal and non-verbal communication etiquette and active listening techniques in various settings	-	-	-	-
PC13. work collaboratively with others in a team	-	-	-	-
<i>Diversity & Inclusion</i>	1	2	-	-
PC14. communicate and behave appropriately with all genders and PwD	-	-	-	-
PC15. escalate any issues related to sexual harassment at workplace according to POSH Act	-	-	-	-
<i>Financial and Legal Literacy</i>	2	3	-	-
PC16. select financial institutions, products and services as per requirement	-	-	-	-
PC17. carry out offline and online financial transactions, safely and securely	-	-	-	-
PC18. identify common components of salary and compute income, expenses, taxes, investments etc	-	-	-	-
PC19. identify relevant rights and laws and use legal aids to fight against legal exploitation	-	-	-	-
<i>Essential Digital Skills</i>	3	4	-	-
PC20. operate digital devices and carry out basic internet operations securely and safely	-	-	-	-
PC21. use e- mail and social media platforms and virtual collaboration tools to work effectively	-	-	-	-
PC22. use basic features of word processor, spreadsheets, and presentations	-	-	-	-



Qualification Pack

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Entrepreneurship</i>	2	3	-	-
PC23. identify different types of Entrepreneurship and Enterprises and assess opportunities for potential business through research	-	-	-	-
PC24. develop a business plan and a work model, considering the 4Ps of Marketing Product, Price, Place and Promotion	-	-	-	-
PC25. identify sources of funding, anticipate, and mitigate any financial/ legal hurdles for the potential business opportunity	-	-	-	-
<i>Customer Service</i>	1	2	-	-
PC26. identify different types of customers	-	-	-	-
PC27. identify and respond to customer requests and needs in a professional manner.	-	-	-	-
PC28. follow appropriate hygiene and grooming standards	-	-	-	-
<i>Getting ready for apprenticeship & Jobs</i>	2	3	-	-
PC29. create a professional Curriculum vitae (Résumé)	-	-	-	-
PC30. search for suitable jobs using reliable offline and online sources such as Employment exchange, recruitment agencies, newspapers etc. and job portals, respectively	-	-	-	-
PC31. apply to identified job openings using offline /online methods as per requirement	-	-	-	-
PC32. answer questions politely, with clarity and confidence, during recruitment and selection	-	-	-	-
PC33. identify apprenticeship opportunities and register for it as per guidelines and requirements	-	-	-	-
NOS Total	20	30	-	-



Qualification Pack

National Occupational Standards (NOS) Parameters

NOS Code	DGT/VSQ/N0102
NOS Name	Employability Skills (60 Hours)
Sector	Cross Sectoral
Sub-Sector	Professional Skills
Occupation	Employability
NSQF Level	4
Credits	2
Version	1.0
Last Reviewed Date	18/02/2025
Next Review Date	18/02/2028
NSQF Clearance Date	18/02/2025

Assessment Guidelines and Assessment Weightage

Assessment Guidelines

Assessment will be based on the concept of Independent Assessors empaneled with West Bengal State Council of Technical & Vocational Education & Skill Development (WBSCT&VE&SD), identified, selected, trained and certified on Assessment techniques. These Assessors would be aligned to assess as per the laid down criteria.

WBSCT&VE&SD would conduct assessment only at the training centers or designated testing centers authorized by WBSCT&VE&SD. Ideally, the assessment will be a continuous process comprising of two distinct steps:

- Continuous assessment by Trainers
- Term end /Final Assessment by WBSCT&VE&SD

Each National Occupational Standard (NOS) in the respective QPs will be assigned weightage. Each Performance Criteria in the NOS will be assigned marks for theory and/or practical based on relative importance and criticality of function. This will facilitate preparation of question bank / paper sets for each of the QPs. Each of these papers sets/question banks created by subject matter experts through WBSCT&VE&SD, especially with regard to the practical test and the defined tolerances, finish, accuracy



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etc. The following tools are proposed to be used for final assessment:

i. Written Test: This will comprise of (i) True/False Statements and/or (ii) Multiple Choice Questions and/or (iii) Matching Type Questions. Online system for this will be preferred.

ii. Practical Test: This will comprise a test job to be prepared as per project briefing following appropriate working steps, using necessary tools, equipment and instruments. Through observation it will be possible to ascertain candidate's aptitude, attention to details, quality consciousness etc.

iii. Structured Viva-voce: This tool will be used to assess the conceptual understanding and the behavioral aspects as regards the job role and the specific task at hand.

Minimum Aggregate Passing % at QP Level : 70

(Please note: Every Trainee should score a minimum aggregate passing percentage as specified above, to successfully clear the Qualification Pack assessment.)

Assessment Weightage

Compulsory NOS

National Occupational Standards	Theory Marks	Practical Marks	Project Marks	Viva Marks	Total Marks	Weightage
WBSC/AGR/N3101. Identify the Scope & importance of Mushroom cultivation in India.	50	40	-	-	90	9
WBSC/AGR/N3102. Prepare & pasteurize the compost necessary to cultivate mushrooms.	24	60	-	-	84	8
WBSC/AGR/N3103. Select commercially important species of mushroom and design appropriate site to cultivate mushrooms.	26	60	-	-	86	9
WBSC/AGR/N3104. Undertake disease control and pest management activities, casing and pinning for mushroom cultivation.	24	60	-	-	84	8



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National Occupational Standards	Theory Marks	Practical Marks	Project Marks	Viva Marks	Total Marks	Weightage
WBSC/AGR/N3105.Demonstrate harvest & post-harvest procedures of mushrooms.	26	140	-	-	166	17
WBSC/AGR/N3106.Work in real job situation with special emphasis on basic safety and hazards in this domain (Mushroom Cultivator).	-	440	-	-	440	44
DGT/VSQ/N0102.Employability Skills (60 Hours)	20	30	-	-	50	5
Total	170	830	-	-	1000	100



Qualification Pack

Acronyms

NOS	National Occupational Standard(s)
NSQF	National Skills Qualifications Framework
QP	Qualifications Pack
TVET	Technical and Vocational Education and Training
NOS	National Occupational Standard(s)
NSQF	National Skills Qualifications Framework
QP	Qualifications Pack
TVET	Technical and Vocational Education and Training
AA	Assessment Agency
AB	Awarding Body
ISCO	International Standard Classification of Occupations
NCO	National Classification of Occupations
NCrF	National Credit Framework
NQR	National Qualification Register



Qualification Pack

Glossary

Sector	Sector is a conglomeration of different business operations having similar business and interests. It may also be defined as a distinct subset of the economy whose components share similar characteristics and interests.
Sub-sector	Sub-sector is derived from a further breakdown based on the characteristics and interests of its components.
Occupation	Occupation is a set of job roles, which perform similar/ related set of functions in an industry.
Job role	Job role defines a unique set of functions that together form a unique employment opportunity in an organisation.
Occupational Standards (OS)	OS specify the standards of performance an individual must achieve when carrying out a function in the workplace, together with the Knowledge and Understanding (KU) they need to meet that standard consistently. Occupational Standards are applicable both in the Indian and global contexts.
Performance Criteria (PC)	Performance Criteria (PC) are statements that together specify the standard of performance required when carrying out a task.
National Occupational Standards (NOS)	NOS are occupational standards which apply uniquely in the Indian context.
Qualifications Pack (QP)	QP comprises the set of OS, together with the educational, training and other criteria required to perform a job role. A QP is assigned a unique qualifications pack code.
Unit Code	Unit code is a unique identifier for an Occupational Standard, which is denoted by an 'N'
Unit Title	Unit title gives a clear overall statement about what the incumbent should be able to do.
Description	Description gives a short summary of the unit content. This would be helpful to anyone searching on a database to verify that this is the appropriate OS they are looking for.
Scope	Scope is a set of statements specifying the range of variables that an individual may have to deal with in carrying out the function which have a critical impact on quality of performance required.
Knowledge and Understanding (KU)	Knowledge and Understanding (KU) are statements which together specify the technical, generic, professional and organisational specific knowledge that an individual needs in order to perform to the required standard.



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Organisational Context	Organisational context includes the way the organisation is structured and how it operates, including the extent of operative knowledge managers have of their relevant areas of responsibility.
Technical Knowledge	Technical knowledge is the specific knowledge needed to accomplish specific designated responsibilities.
Core Skills/ Generic Skills (GS)	Core skills or Generic Skills (GS) are a group of skills that are the key to learning and working in today's world. These skills are typically needed in any work environment in today's world. These skills are typically needed in any work environment. In the context of the OS, these include communication related skills that are applicable to most job roles.
Electives	Electives are NOS/set of NOS that are identified by the sector as contributive to specialization in a job role. There may be multiple electives within a QP for each specialized job role. Trainees must select at least one elective for the successful completion of a QP with Electives.
Options	Options are NOS/set of NOS that are identified by the sector as additional skills. There may be multiple options within a QP. It is not mandatory to select any of the options to complete a QP with Options.
Sector	Sector is a conglomeration of different business operations having similar business and interests. It may also be defined as a distinct subset of the economy whose components share similar characteristics and interests.
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Unit Code	Unit code is a unique identifier for an Occupational Standard, which is denoted by an N
Unit Title	Unit title gives a clear overall statement about what the incumbent should be able to do.
Description	Description gives a short summary of the unit content. This would be helpful to anyone searching on a database to verify that this is the appropriate OS they are looking for.
Scope	Scope is a set of statements specifying the range of variables that an individual may have to deal with in carrying out the function which have a critical impact on quality of performance required.
Knowledge and Understanding	Knowledge and Understanding (KU) are statements which together specify the technical, generic, professional and organisational specific knowledge that an individual needs in order to perform to the required standard.
Organisational Context	Organisational context includes the way the organisation is structured and how it operates, including the extent of operative knowledge managers have of their relevant areas of responsibility
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Options	Options are NOS/set of NOS that are identified by the sector as additional skills. There may be multiple options within a QP. It is not mandatory to select any of the options to complete a QP with Options.
Qualification	A formal outcome of an assessment and validation process which is obtained when a competent body determines that an individual has achieved learning outcomes to given standards.
Qualification File	A Qualification File is a template designed to capture necessary information of a Qualification from the perspective of NSQF compliance. The Qualification File will be normally submitted by the awarding body for the qualification.
Long Term Training	Long-term skilling means any vocational training program undertaken for a year and above. https://ncvet.gov.in/sites/default/files/NCVET.pdf .